

原力覺醒 2025年電機資訊學院 專題暨創意構想競賽

專題
製作

Team Name：草莓不想排隊

編號：

Title：Food Image Recognition and RAG-Integrated
VLM Nutrition Recommendation Model

展演語言：

Team Member：陳瀚傑、王筠翔、李宗翰

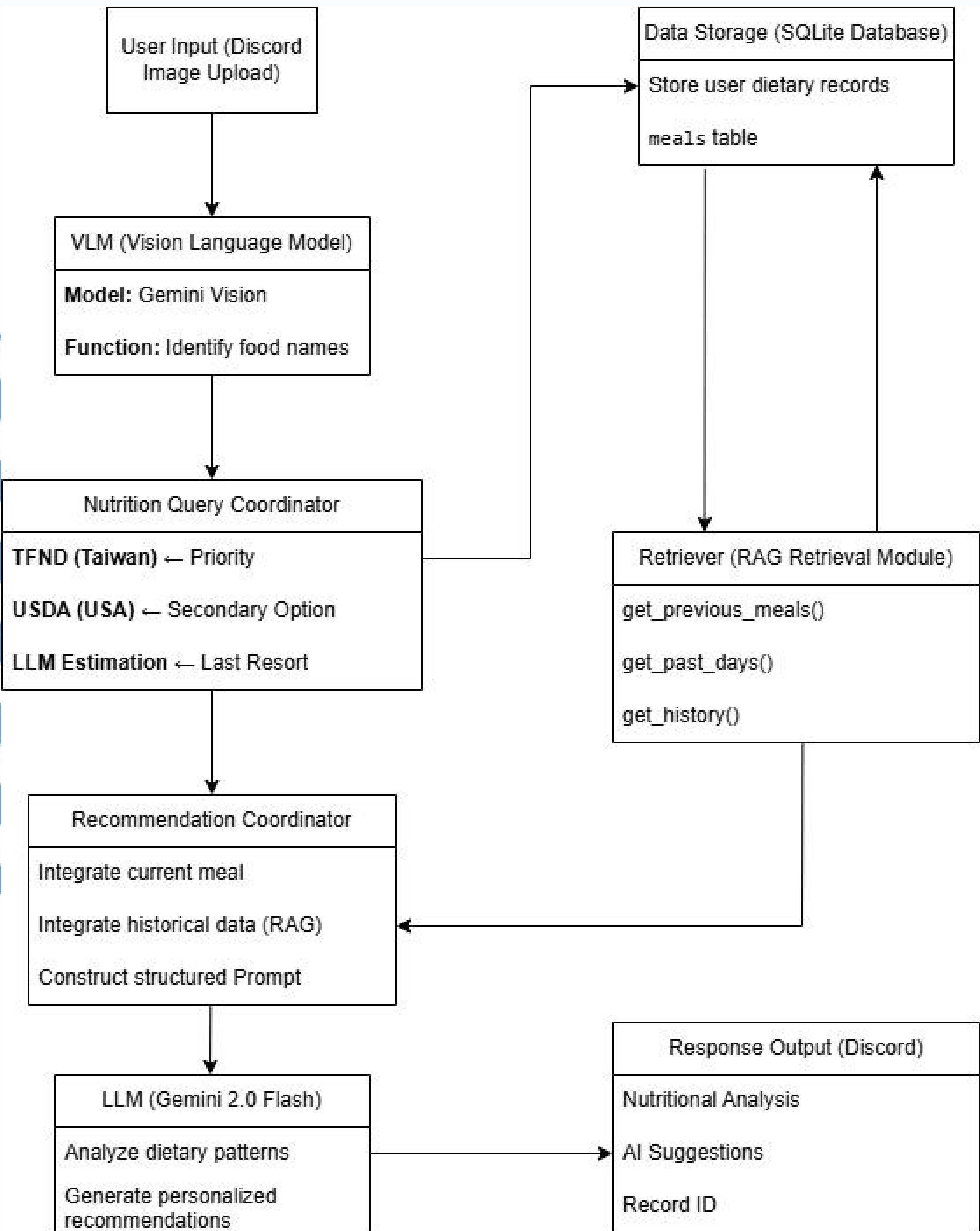
Abstract

This project builds an intelligent dietary recommendation system that integrates food image recognition, RAG, and LLM. Users upload meal photos via a Discord Bot, and a VLM identifies the food and retrieves nutritional data from Taiwan’s Ministry of Health and Welfare database and the USDA API. RAG then combines the new entry with past records to generate a structured nutritional summary. Finally, the LLM provides personalized dietary analysis, health reminders, and meal recommendations based on the user’s goals and history.

Motivation

As health awareness grows, traditional diet-tracking apps remain inconvenient due to manual input and subscription fees. Advances in AI and LLMs enable automatic nutrition analysis from food images. This project builds an “AI Food Nutritionist” system that integrates food image recognition, RAG, and LLM to provide instant nutritional insights and personalized recommendations simply by uploading a photo.

Methodology



Results

食物營養師 應用 2025/11/26 下午 03:27

✓ 飲食分析完成！

🕒 餐次資訊

🌙 晚餐

🔍 識別結果

小籠包 (Xiao Long Bao/Soup Dumplings) - 6、凉拌猪耳朵 (Cold Pig Ear Salad)、凉拌黄瓜 (Cucumber Salad)、烧卖 (Shaomai/Siumai) - 2、海带丝 (Shredded Kelp/Seaweed Salad)、汤面 (Soup noodles)、葱花 (Chopped Scallions/Green Onions)

📊 營養分析

- 小籠包 (Xiao Long Bao/Soup Dumplings) - 6: 323 kcal
- 凉拌猪耳朵 (Cold Pig Ear Salad): 236 kcal
- 凉拌黄瓜 (Cucumber Salad): 34 kcal
- 烧卖 (Shaomai/Siumai) - 2: 50 kcal
- 海带丝 (Shredded Kelp/Seaweed Salad): 45 kcal
- 汤面 (Soup noodles): 53 kcal
- 葱花 (Chopped Scallions/Green Onions): 31 kcal

📋 總營養素

🔥 熱量: 772 kcal

🤖 AI 個人化建議

- 立刻停止進食高熱量食物，多喝水增加飽足感。
- 下一餐以高纖維、低熱量的蔬菜水果為主，例如沙拉或清炒蔬菜。
- 注意隱藏熱量來源，如沾醬、湯底、油炸物等，減少食用。

📄 記錄資訊

使用 /history 查看完整記錄



💡 保持健康飲食，持續追蹤您的營養攝取 (本系統所產生之結果僅供參考)
• 2025/11/26 下午 03:27